



MANASWI BICHALA

B.Tech. - Computer Science and Engineering

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PROFESSIONAL SUMMARY

Computer Science undergraduate with a strong foundation in programming, data structures, algorithms, object-oriented programming, and software development. Passionate about building efficient, user-focused applications and continuously expanding technical expertise through hands-on projects. Quick to learn new technologies, adaptable to dynamic environments, and committed to writing clean, maintainable code. Eager to contribute to collaborative teams while gaining practical industry experience and delivering impactful software solutions.

KEY EXPERTISE

C Java OOPs Data Structures Operating Systems Computer Networking Image Processing
Image Segmentation Computer Vision Pattern Recognition Python Github SQL GIT OpenCV
Problem Solving Analytical Skills Pandas Matplotlib Visual Studio

EDUCATION

G. Narayanamma Institute of Technology and Science

2023 - 2027

B.Tech. - Computer Science and Engineering | CGPA: 8 / 10

Sri Chaitanya Junior College, Hayathnagar

2023

12th | Telangana State Board | Percentage: 97.60 / 100

MJPTBCWR(G) School

2021

10th | SSC | CGPA: 10 / 10

PROJECTS

Land Use Land Cover (LULC) Classification using Deep Learning

Key Skills: Python Deep Learning ResNet50 Computer Vision TensorFlow Remote Sensing Image Processing

Developed an automated system to classify land use and land cover using satellite imagery and a ResNet50-based deep learning model. Trained the model on the EuroSAT dataset to categorize regions into classes such as urban areas, vegetation, water bodies, agricultural land, and barren land. Implemented tile-based image processing to improve spatial pattern detection and generate color-coded LULC maps for environmental monitoring and urban planning.

Para Detect AI-based Malaria Detection System

Key Skills: Python Opencv Deep Learning Computer Vision streamlit

Developed a deep learning-based system to detect malaria parasites from microscopic blood smear images. Built a web interface using Streamlit that allows users to upload cell images and receive predictions as Parasitized or Uninfected. Used Python, TensorFlow/Keras, OpenCV, and image preprocessing techniques to train and deploy the model for quick and accessible malaria diagnosis.

Real Time Violence Detection Using Deep Learning

Key Skills: Python Opencv Computer Vision Deep Learning cnn cbam Image Processing

Developed an intelligent violence detection system that analyzes surveillance video frames to classify violent and non-violent activities using a Convolutional Neural Network (CNN) enhanced with a Convolutional Block Attention Module (CBAM). Implemented video frame extraction, preprocessing, and data augmentation using OpenCV, and trained the model with TensorFlow/Keras to improve feature representation and detection accuracy. The system is designed to support real-time surveillance by generating alerts upon detecting violent incidents, reducing manual monitoring efforts and enabling faster response in public safety applications.

PUBLICATIONS / RESEARCH / WHITE PAPERS

Classifying Land Use and Land Cover for Sustainable Urban Planning and Ecosystem Conservation 11 Jun, 2025

Mentor: Dr. K.L.S.Soujanya
Accurate classification of Land Use and Land Cover (LULC) is fundamental to understanding the spatial distribution of natural and anthropogenic features on the Earth's surface. It provides essential insights for urban planning, agricultural development, environmental monitoring, and resource management. The rapid pace of urbanization—particularly in developing regions—has amplified the demand for timely and precise LULC data. Traditional methods, such as manual interpretation and field surveys, are increasingly inadequate due to limitations in scalability, efficiency, and consistency. This study proposes an automated LULC classification approach that leverages deep learning and remote sensing technologies. Utilizing the ResNet50 deep convolutional neural network and the EuroSAT dataset comprising multispectral satellite imagery, the model is trained to classify land cover types such as urban areas, vegetation, water bodies, agricultural zones, and barren land. The classification process involves tiling satellite images into smaller segments, enabling fine-grained spatial pattern detection and high-resolution mapping.

ASSESSMENTS / CERTIFICATIONS

Generative AI in Action
Provider: IBM

AI Tools Workshop
Provider: be10x

PERSONAL INTERESTS / HOBBIES

- Listening to music
- Reading books
- Yoga

WEB LINKS

- Github - <https://github.com/manaswi-7>
- Other - <https://leetcode.com/u/Lr7Vdo8yWA/>

PERSONAL DETAILS

Gender: Female
Marital Status: Single
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Date of Birth: 21 Feb, 2006
Known Languages: English, Telugu, Hindi, Spanish(Intermediate)
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